

Unit - II part - A

Date : _____

$\vec{A}$ = vector

$\vec{E}$ = Electric field vector

$\vec{B}$ = Magnetic field vector

dot product = $\vec{A} \cdot \vec{B}$

Cross product = $\vec{A} \times \vec{B}$

$\vec{A} = A\hat{i} + B\hat{j} + C\hat{k}$

$\hat{i}, \hat{j}, \hat{k}$ = unit vectors

$$\hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1$$

$$\hat{i} \cdot \hat{j} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0$$

$$\hat{i} \times \hat{i} = 0, \hat{j} \times \hat{j} = 0, \hat{k} \times \hat{k} = 0$$

$$\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$$

$$\hat{j} \times \hat{i} = -\hat{k}, \hat{k} \times \hat{j} = -\hat{i}, \hat{i} \times \hat{k} = -\hat{j}$$

Unit

∇ = del operator

$$\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

Divergence of any vector field $\vec{A}$

(D - Divergence,
D - Dot product)

$$\vec{\nabla} \cdot \vec{A} = \text{divergence of } \vec{A}$$

curl of any vector field $\vec{A}$

(C - curl
C - cross product)

$$\vec{\nabla} \times \vec{A} = \text{curl of } \vec{A}$$

$\vec{E}$ = Electric field

divergence of electric field $\vec{E} = \vec{\nabla} \cdot \vec{E}$

curl of electric field $\vec{E} = \vec{\nabla} \times \vec{E}$

divergence of magnetic field $\vec{B} = \vec{\nabla} \cdot \vec{B}$

curl of magnetic field $\vec{B} = \vec{\nabla} \times \vec{B}$

$\therefore \int dx = \text{line integration}$

$\iint dx dy = \text{surface integration}$

$\iiint dx dy dz = \text{volume integration}$

Gauss Divergence Theorem:

→ According to Gauss divergence theorem, the surface integration of a vector field ($\vec{A}$) over a close path is equal to volume integration of divergence of that field.

$$\oiint_S \vec{A} \cdot d\vec{s} = \iiint_V \vec{\nabla} \cdot \vec{A} dv$$

$dA = ds = \text{small surface area}$

$dv = \text{small volume element}$

Stoke's Theorem

→ According to Stoke's theorem, the line integration of a vector field $\vec{A}$ over a closed path is equal to surface integration of curl of that field.

$$\oint \vec{A} \cdot d\vec{l} = \iint_S \nabla \times \vec{A} \cdot d\vec{s}$$

$d\vec{l}$ = small line element

$d\vec{s}$ = small surface area

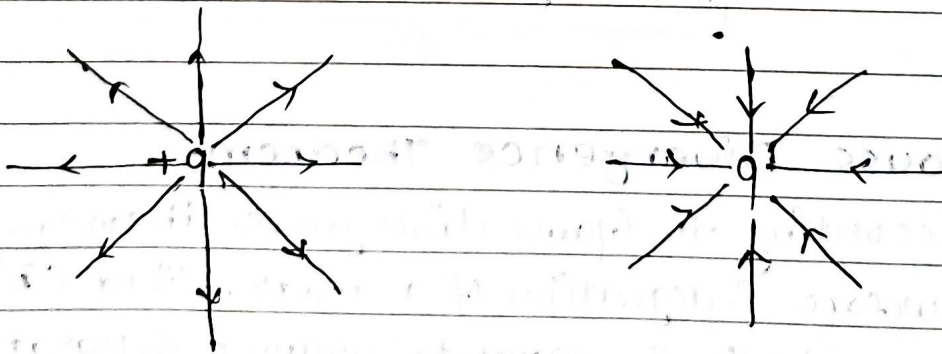
Electric field

→ The region around the source charge up to which the test charge Q can experience the force exerted by source charge is called electric field.

→ Electric field $\vec{E} = \lim_{Q \rightarrow 0} \frac{\vec{F}}{Q}$

$$\vec{F} = \frac{K q_1 q_2}{r^2}$$

Physical meaning of divergence



→ Divergence of electric field $\vec{E}$ tells us the flow of electric field lines inward / outward from of any surface S .

Divergence = outward / inward flow

$$\text{Divergence of Electric field } \vec{E} = \nabla \cdot \vec{E}$$

- Charges at rest produce electric field
- Charges in motion produce magnetic field.

$V = \text{Scalar}$

$$\vec{\nabla} V = \text{gradient of } V = \text{grad}(V)$$

Gauss law in electrostatic (in Integral form)

- The electric flux through a closed surface S is equal to $\frac{1}{\epsilon_0}$ times the total charge enclosed in it.

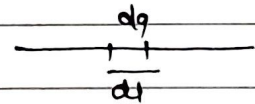
$$\text{Electric flux} = \oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0} \quad \begin{array}{l} \text{Integral form} \\ \text{of Gauss law} \end{array}$$

* Line Charge

- Line charge density λ

$$\lambda = \frac{\text{Charge}}{\text{length}} = \frac{dq}{dl}$$

$$dq = \lambda dl \Rightarrow \boxed{q = \int \lambda dl}$$

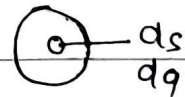


* Surface Charge

surface charge density σ

$$\sigma = \frac{\text{Charge}}{\text{Area}} = \frac{dq}{ds}$$

$$dq = \sigma ds \Rightarrow \boxed{q = \iint \sigma ds}$$

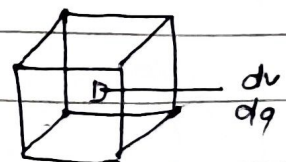


* Volume Charge

volume charge density ρ

$$\rho = \frac{\text{Charge}}{\text{volume}} = \frac{dq}{dv}$$

$$\Rightarrow dq = \rho dv \Rightarrow \boxed{q = \iiint \rho dv}$$



Divergence of electric field $\vec{E}$ ($\nabla \cdot \vec{E}$)

→ The integral form of Gauss's law.

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0} \quad \text{--- (1)}$$

using Gauss Divergence Theorem, L.H.S becomes

$$\iiint_V \nabla \cdot \vec{E} \, dv = \frac{1}{\epsilon_0} \iiint_V \rho \, dv$$

$$\iiint_V \nabla \cdot \vec{E} \, dv - \iiint_V \frac{\rho}{\epsilon_0} \, dv = 0 \quad \left[\because \rho = \text{volume charge density} \right]$$

$$\iiint_V \left(\nabla \cdot \vec{E} - \frac{\rho}{\epsilon_0} \right) dv = 0$$

Since $dv \neq 0$

$$\nabla \cdot \vec{E} - \frac{\rho}{\epsilon_0} = 0$$

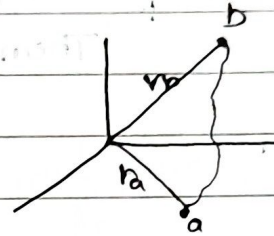
$$\Rightarrow \boxed{\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}} = \text{div } \vec{E}$$

This is differential form of Gauss law.

Curl of Electric field $\vec{E}$ ($\nabla \times \vec{E}$)

→ The electric field produced by a charge q is

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$



Take line integral both sides

$$\int_a^b \vec{E} \cdot d\vec{l} = \int_a^b \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dl = \frac{q}{4\pi\epsilon_0} \int_a^b \frac{1}{r^2} dl$$

$$= \frac{q}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_a^b = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_b} - \frac{1}{r_a} \right]$$

for closed path

$$r_a = r_b$$

$$\oint \vec{E} \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_a} - \frac{1}{r_a} \right]$$

$$\oint \vec{E} \cdot d\vec{l} = 0$$

Using Stoke's Theorem L.H.S

$$\iint_S (\nabla \times \vec{E}) \cdot d\vec{s} = 0$$

$$d\vec{s} \neq 0$$

$$\Rightarrow \boxed{\nabla \times \vec{E} = 0}$$

$\times$ = cross product

$\oint$ = circulation

$\nabla \times$ - curl

→ It means that electric field lines do not form circular loops.

'curl' means = Circulation

~~then~~ Since,

$$\vec{\nabla} \times \vec{E} = 0$$

$$\text{Then, } \boxed{\vec{E} = -\vec{\nabla} v}$$

, v = scalar potential

$$\vec{E} = -\text{grad}(v)$$

$$\vec{E} = -\text{gradient of } v$$

V.I

4 marks

Laplace Theorem and Poisson Theorem.

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \text{--- (1)}$$

$$\vec{E} = -\vec{\nabla} v \quad \text{--- (2)}$$

Put (2) in (1)

$$\vec{\nabla} \cdot (-\vec{\nabla} v) = \frac{\rho}{\epsilon_0}$$

$$\boxed{\nabla^2 v = -\frac{\rho}{\epsilon_0}}$$

Poisson
Theorem

(3)

ρ = volume charge density

$\vec{E}$ = electric field

$\vec{\nabla}$ = del operator

v = scalar ~~prod~~
potential

In charge free region

charge density $\Rightarrow \rho = 0$

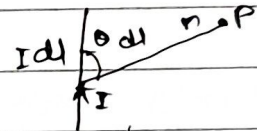
Put $\rho = 0$ in eq (3)

$$\boxed{\nabla^2 v = 0} \text{ --- Laplace Theorem}$$

Magnetic field

→ The region around a current carrying conductor where the magnetic effects can be experienced is called magnetic field.

Biot-Savart Law



→ According to Biot-Savart Law, dB = Small magnetic field.

$$dB \propto Idl$$

$$\propto \sin \theta$$

$$\propto \frac{1}{r^2}$$

$$dB \propto \frac{Idl \sin \theta}{r^2}$$

$$dB = k \frac{Idl \sin \theta}{r^2}$$

$$\therefore k = \frac{\mu_0}{4\pi}$$

$$dB = \frac{\mu_0 Idl \sin \theta}{4\pi r^2} \hat{r}$$

$$\hat{r} = \frac{\vec{r}}{r}$$

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \vec{r}}{r^3}$$

$$\boxed{\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l} \times \vec{r}}{r^3}}$$

Ampere's Circuital law

→ Integral form of Ampere's Circuital law.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

 μ_0 = permeability of free space. $d\vec{l}$ = line element. $d\vec{s}$ or $d\vec{A}$ = surface area dv = volume of element

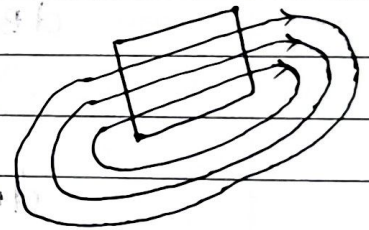
→ The line integration of magnetic field over a closed path is equal to μ_0 times the current flowing through it.

Divergence of Magnetic field $\vec{B}$ ($\nabla \cdot \vec{B}$)

→ The Gauss law in Magnetostatics:

$$\oiint \vec{B} \cdot d\vec{s} = \text{Magnetic flux}$$

$$\oiint \vec{B} \cdot d\vec{s} = 0$$



→ Since magnetic lines of force form circular loops, so, the number of magnetic lines leaving the surface is equal to number of magnetic lines entering a surface. So, total magnetic flux

$$\oiint \vec{B} \cdot d\vec{s} = 0$$

Using Gauss Divergence Theorem.

$$\iiint \nabla \cdot \vec{B} \, dv = 0$$

Since, $du \neq 0$

$$\Rightarrow \boxed{\nabla \cdot \vec{B} = 0}$$

* Physical meaning

- magnetic monopoles do not exist.
- Magnetic lines of force always form circular loops.

Curl of magnetic field $\vec{B}$ ($\nabla \times \vec{B}$)

→ The Ampere Circuital law in integral form is

$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 I \quad \text{--- (1)}$$

$I = \text{current}$

using Stoke's Theorem

$$\oint_L \vec{B} \cdot d\vec{l} = \iint_S \nabla \times \vec{B} \cdot d\vec{s}$$

Put this value in L.H.S of eq (1)

$$\iint_S \nabla \times \vec{B} \cdot d\vec{s} = \mu_0 I \quad \text{--- (2)}$$

We know that the current density $\vec{J}$ for small current dI through a small surface area element ds is current density ($\vec{J}$) =

$$\vec{J} = \frac{\text{current}}{\text{Area}} = \frac{dI}{ds}$$

$$dI = \vec{J} \cdot d\vec{s}$$

$$I = \iint_S \vec{J} \cdot d\vec{s}$$

Put the value in R.H.S of eq (2)

$$\iint_S \vec{\nabla} \times \vec{B} \cdot d\vec{s} = \mu_0 \iint_S \vec{J} \cdot d\vec{s} = \iint_S \mu_0 \vec{J} \cdot d\vec{s}$$

$$\iint_S (\vec{\nabla} \times \vec{B} - \mu_0 \vec{J}) \cdot d\vec{s} = 0$$

Since, $d\vec{s} = 0$ small area element.

$$\vec{\nabla} \times \vec{B} - \mu_0 \vec{J} = 0$$

$$\boxed{\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}} \rightarrow \text{curl of } \vec{B}$$

↳ This is the differential form of
Ampere Circuital law.

V.V.V.
6marks

→ Calculation of vector potential ($\vec{A}$) for a given magnetic field by using Stoke's Theorem.

→ we know that from Biot - Savart laws, magnetic field due to a current loop is given by,

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_L \frac{d\vec{u} \times \vec{r}}{r^3} \quad \text{--- (1)}$$

we know that $\frac{\vec{r}}{r^3} = -\vec{\nabla} \left(\frac{1}{r} \right)$ --- To remember

$$\vec{B} = -\frac{\mu_0}{4\pi} \oint_L d\vec{u} \times \left[\vec{\nabla} \left(\frac{1}{r} \right) \right]$$

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_L \vec{\nabla} \left(\frac{1}{r} \right) \times d\vec{u} \quad \text{--- (2)} \quad [\because \vec{A} \times \vec{B} = -\vec{B} \times \vec{A}]$$

$\therefore \frac{1}{r} = \text{scalar}$

$\therefore d\vec{u} = \text{vector}$

$$\vec{\nabla} \times (p\vec{a}) = p \vec{\nabla} \times \vec{a} - \vec{a} \times \underline{\vec{\nabla} p}$$

$$\vec{\nabla} \times \left(\frac{1}{r} d\vec{u} \right) = \frac{1}{r} \vec{\nabla} \times d\vec{u} - d\vec{u} \times \underline{\vec{\nabla} \left(\frac{1}{r} \right)}$$

$$\boxed{\vec{\nabla} \times \left(\frac{1}{r} d\vec{u} \right) = \frac{1}{r} \vec{\nabla} \times d\vec{u} + \vec{\nabla} \left(\frac{1}{r} \right) \times d\vec{u} \quad [\because \vec{A} \times \vec{B} = -\vec{B} \times \vec{A}]}$$

$$\Rightarrow \vec{\nabla} \left(\frac{1}{r} \right) \times d\vec{u} = \vec{\nabla} \times \left(\frac{1}{r} d\vec{u} \right) - \frac{1}{r} \vec{\nabla} \times d\vec{u}$$

Put this value in eq (2)

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_L \vec{\nabla} \times \left(\frac{1}{r} d\vec{l} \right) - \frac{1}{r} \underbrace{\vec{\nabla} \times d\vec{l}}_0$$

$$\therefore \boxed{\vec{\nabla} \times d\vec{l} = 0}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_L \vec{\nabla} \times \left(\frac{1}{r} d\vec{l} \right)$$

$$\vec{B} = \oint_L \vec{\nabla} \times \left(\frac{\mu_0}{4\pi} \frac{1}{r} d\vec{l} \right)$$

$$\vec{B} = \vec{\nabla} \times \oint_L \frac{\mu_0}{4\pi} \frac{1}{r} d\vec{l} \quad \text{----- (3)}$$

Since,

$$\boxed{\vec{B} = \vec{\nabla} \times \vec{A}} \quad \text{----- (4)}$$

Comparing (3) and (4)

$$\vec{A} = \oint_L \frac{\mu_0}{4\pi} \frac{1}{r} d\vec{l}$$

$$\boxed{\begin{array}{l} \vec{\nabla} \cdot \vec{B} = 0 \\ \hookrightarrow \text{then } \vec{B} = \vec{\nabla} \times \vec{A} \end{array}}$$

Ques. Prove that $\vec{\nabla} \times \vec{d} = 0$

$$\vec{d} = \hat{i} dx + \hat{j} dy + \hat{k} dz$$

$$\vec{\nabla} = \hat{i} \frac{d}{dx} + \hat{j} \frac{d}{dy} + \hat{k} \frac{d}{dz}$$

$$\vec{\nabla} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ dx & dy & dz \end{vmatrix}$$

$$= \hat{i} \left(\frac{dz}{dy} - \frac{dy}{dz} \right) + \hat{j} \left(\frac{dx}{dz} - \frac{dz}{dx} \right) + \hat{k} \left(\frac{dy}{dx} - \frac{dx}{dy} \right)$$

$$= \hat{i} (0 - 0) + \hat{j} (0 - 0) + \hat{k} (0 - 0)$$

$$= (\hat{i} \times 0) + (\hat{j} \times 0) + (\hat{k} \times 0)$$

$$= 0 + 0 + 0$$

$$= 0$$

Unit-2 part-B

Date: _____

Electromagnetic waves

Equation of continuity

Physical meaning:

$$\vec{\nabla} \cdot \vec{J} + \frac{d\rho}{dt} = 0$$

$\vec{J}$ = current density

ρ = volume charge density.

This equation tells us about the conservation of charge.

current $= I$ = flow of charge

$$I = \frac{-dq}{dt} \quad \text{--- (1)}$$

we know that, if ρ is volume charge density, then current in volume V is

$$q = \iiint_V \rho \, dv \quad \text{--- (2)}$$

$$q = \iiint_V \rho \, dv$$

$$I = \iint_S \sigma \, ds$$

$$q = \int I \, dt$$

$$\text{current } \vec{J} = \frac{\text{current}}{\text{Area}} = \frac{dI}{ds}$$

$$dI = \vec{J} \cdot d\vec{s}$$

$$I = \oint_S \vec{J} \cdot d\vec{s} \quad \text{--- (3)}$$

Put values from (2) and (3) in (1)

$$\oint_S \vec{J} \cdot d\vec{s} = -\frac{q}{dt} \iiint_V \rho dv$$

using Gauss Divergence Theorem L.H.S becomes

$$\iiint_V \vec{\nabla} \cdot \vec{J} dv = -\iiint_V \frac{d\rho}{dt} dv$$

$$\iiint_V \vec{\nabla} \cdot \vec{J} dv + \iiint_V \frac{d\rho}{dt} dv = 0$$

$$\iiint_V \left(\vec{\nabla} \cdot \vec{J} + \frac{d\rho}{dt} \right) dv = 0$$

Since $dv \neq 0$

$$\boxed{\vec{\nabla} \cdot \vec{J} + \frac{d\rho}{dt} = 0}$$

This is equation of continuity.

Unit
(6 mark)

Maxwell's Equation in Differential form are :

① $\vec{\nabla} \cdot \vec{D} = \rho$, $\vec{D} = \epsilon_0 \vec{E}$ = Gauss's law in electrostatic

②. $\vec{\nabla} \cdot \vec{B} = 0$ Gauss's law in Magnetostatic

$\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$ - Faraday's law of Electromagnetic Induction

$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$, $\vec{B} = \mu_0 \vec{H}$ Modified Ampere
Circular law

Proof of Maxwell's Equation differential form,

①. $\boxed{\vec{\nabla} \cdot \vec{D} = \rho} \rightarrow \text{To Prove}$

Proof: The Gauss's law in Electrostatic is

$$\oint \vec{E} \cdot d\vec{s} = \frac{\rho q}{\epsilon_0}$$

$$\oint (\epsilon_0 \vec{E}) \cdot d\vec{s} = \rho q$$

$$\oint \vec{D} \cdot d\vec{s} = \rho q$$

Using Gauss's Divergence theorem in L.H.S

$$\iiint \vec{\nabla} \cdot \vec{D} \, dv = q$$

$$\text{Put } q = \iiint \rho \, dv$$

$$\iiint_V \vec{\nabla} \cdot \vec{D} \, dv = \iiint_V \rho \, dv$$

$$\iiint_V (\vec{\nabla} \cdot \vec{D} - \rho) \, dv = 0$$

$$dv \neq 0$$

$$\vec{\nabla} \cdot \vec{D} - \rho = 0$$

$$\boxed{\vec{\nabla} \cdot \vec{D} = \rho}$$

Proved

②. Maxwell second Equation.

$$\boxed{\vec{\nabla} \cdot \vec{B} = 0} \rightarrow \text{Prove}$$

Proof: Gauss law in magnetostatic is

$$\oint_S \vec{B} \cdot d\vec{s} = 0$$

Using Gauss Divergence Theorem.

$$\iiint_V (\vec{\nabla} \cdot \vec{B}) \, dv = 0$$

$$dv \neq 0$$

$$\boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

Proved

Maxwell's third Equation

$$\vec{\nabla} \times \vec{E} = - \frac{d\vec{B}}{dt} \longrightarrow \text{To prove}$$

Proof :-

According to Faraday's law of electromagnetic induction, the e.m.f is induced given by

$$e = - \frac{d\phi}{dt}, \quad \phi = \text{magnetic flux.} \quad \text{--- (1)}$$

$$\text{magnetic flux } \phi = \iint \vec{B} \cdot d\vec{s}$$

Put this value in eq (1)

$$e = - \frac{d}{dt} \iint \vec{B} \cdot d\vec{s}$$

$$e = - \iint \frac{d\vec{B}}{dt} \cdot d\vec{s} \quad \text{--- (2)}$$

The e.m.f is defined as the work done to take a unit positive charge around a closed loop.

$$e = \oint \vec{E} \cdot d\vec{l} \quad \text{--- (3)}$$

Put this value in L.H.S of eq (2)

$$\oint \vec{E} \cdot d\vec{l} = - \iint \frac{d\vec{B}}{dt} \cdot d\vec{s}$$

using Stoke's Theorem in L.H.S

$$\iint_S \vec{\nabla} \times \vec{E} \cdot d\vec{s} = - \iint_S \frac{d\vec{B}}{dt} \cdot d\vec{s}$$

$$\iint_S \vec{\nabla} \times \vec{E} \cdot d\vec{s} + \iint_S \frac{d\vec{B}}{dt} \cdot d\vec{s} = 0$$

$$\iint_S \left(\vec{\nabla} \times \vec{E} + \frac{d\vec{B}}{dt} \right) \cdot d\vec{s} = 0$$

since, $d\vec{s} \neq 0$

$$\vec{\nabla} \times \vec{E} + \frac{d\vec{B}}{dt} = 0$$

$$\boxed{\vec{\nabla} \times \vec{E} = - \frac{d\vec{B}}{dt}}$$

Hence, Proved

Maxwell's fourth Equation

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt} \quad \text{To prove}$$

Proof:-

for time independent currents, the differential form of Ampere's circuital law is

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

$$\vec{B} = \mu_0 \vec{H}$$

$$\boxed{\frac{\vec{B}}{\mu_0} = \vec{H}}$$

$$\vec{\nabla} \times \frac{\vec{B}}{\mu_0} = \vec{J}$$

$$\boxed{\vec{\nabla} \times \vec{H} = \vec{J}}$$

According to modified Ampere's circuital law,
for time dependent currents, we have

$$\vec{\nabla} \times \vec{H} = \vec{J} + \vec{J}' \quad \text{--- (1)} \quad \left| \begin{array}{l} J' = \frac{dD}{dt} \\ \vec{\nabla} \times \vec{H} = \vec{J} + \frac{dD}{dt} \end{array} \right.$$

According to Equation of continuity

$$\boxed{\vec{\nabla} \cdot \vec{J} + \frac{d\rho}{dt} = 0}$$

1 marks

ρ = charge density

$\vec{J}$ = current density

Proved

$$\vec{\nabla} \cdot \vec{J} = -\frac{d\rho}{dt} \quad \text{--- (2)}$$

When divergence of eq (1) both sides

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot (\vec{J} + \vec{J}') \quad \text{--- (3)}$$

Since,

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = 0$$

$$\vec{\nabla} \cdot (\vec{J} + \vec{J}') = 0$$

$$\vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \vec{J}' = 0$$

using eq (2)

$$-\frac{d\rho}{dt} + \vec{\nabla} \cdot \vec{J}' = 0$$

$\therefore J' = \text{Time varying current density}$

$$\vec{\nabla} \cdot \vec{J} = \frac{d\rho}{dt} \quad \text{--- (4)}$$

Now,

Maxwell first equation is

$$\vec{\nabla} \cdot \vec{D} = \rho$$

Put $\rho = \vec{\nabla} \cdot \vec{D}$ in R.H.S of eq (4)

$$\vec{\nabla} \cdot \vec{J} = \frac{d\rho}{dt} = \frac{d(\vec{\nabla} \cdot \vec{D})}{dt}$$

$$\vec{\nabla} \cdot \vec{J} = \vec{\nabla} \cdot \left(\frac{d\vec{D}}{dt} \right)$$

$$\Rightarrow \vec{J} = \frac{d\vec{D}}{dt} \quad \rightarrow \text{Put this value in eq (1)}$$

$$\boxed{\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}}$$

Hence, Proved.

Maxwell's Equations in differential form:

(i) $\vec{\nabla} \cdot \vec{D} = \rho$ $\vec{D} = \epsilon_0 \vec{E} \rightarrow$ Gauss law in electrostatic.

(ii) $\vec{\nabla} \cdot \vec{B} = 0$ $\rightarrow$ Gauss law in magnetostatic

(iii) $\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$ $\rightarrow$ Faraday's law of electromagnetic induction.

(iv) $\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$, $\vec{B} = \mu_0 \vec{H} \rightarrow$ Modified form of Ampere circuital law.

In free space or vacuum

$$\rho = 0, \quad J = 0$$

$$(i) \quad \vec{\nabla} \cdot \vec{D} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$$

$$\vec{\nabla} \times \vec{H} = \frac{d\vec{D}}{dt}$$

In conducting medium.

$$\vec{J} = \sigma \vec{E} \quad \sigma = \text{conductivity.}$$

$$\vec{J} \neq 0$$

6 marks

Maxwell's Equation in Integral form:

(i) Gauss law in electrostatic

$$\text{To Prove} - \oint \vec{D} \cdot d\vec{s} = q, \quad \vec{D} = \epsilon_0 \vec{E}$$

Proof:-

The differential form of Gauss's law or Maxwell's first equation is

$$\vec{\nabla} \cdot \vec{D} = \rho$$

Take volume integration both sides.

$$\iiint_V \vec{\nabla} \cdot \vec{D} \, dv = \iiint_V \rho \, dv \quad \text{--- (1)}$$

using Gauss divergence theorem

$$\iiint_V \vec{\nabla} \cdot \vec{D} \, dv = \oiint_S \vec{D} \cdot d\vec{s}$$

$$q = \iiint_V \rho \, dv$$

Put this value in eq (1)

$$\boxed{\oiint_S \vec{D} \cdot d\vec{s} = q} \quad \text{Integral form of Gauss law in electrostatics.}$$

(EE) Gauss law in Magnetostatic

$$\boxed{\oiint_S \vec{B} \cdot d\vec{s} = 0} \quad \text{— To prove}$$

Proof:-

Maxwell's second equation in differential form is

$$\vec{\nabla} \cdot \vec{B} = 0$$

Take volume integration both sides.

$$\iiint_V \vec{\nabla} \cdot \vec{B} \, dv = 0$$

using Gauss divergence theorem.

$$\iiint_V \vec{\nabla} \cdot \vec{B} \, dv = \oiint_S \vec{B} \cdot d\vec{s}$$

$$\boxed{\oiint_S \vec{B} \cdot d\vec{s} = 0} \quad \text{proved}$$

(iii) Maxwell's Third Equation in Integral form.

To Prove - $\oint_L \vec{E} \cdot d\vec{r} = -\frac{d}{dt} \iint_S \vec{B} \cdot d\vec{s}$

Proof:

Maxwell's third equation in differential form

$$\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$$

Take surface integration both sides.

$$\iint_S \vec{\nabla} \times \vec{E} \cdot d\vec{s} = -\iint_S \frac{d\vec{B}}{dt} \cdot d\vec{s}$$

using Stoke's theorem in L.H.S

$$\oint_L \vec{E} \cdot d\vec{r} = -\iint_S \frac{d\vec{B}}{dt} \cdot d\vec{s}$$

$$\boxed{\oint_L \vec{E} \cdot d\vec{r} = -\frac{d}{dt} \iint_S \vec{B} \cdot d\vec{s}}$$

Proved

(iv) Maxwell's fourth Equation in Integral form

$$\oint_L \vec{H} \cdot d\vec{r} = \iint_S \left(\vec{J} + \frac{d\vec{D}}{dt} \right) \cdot d\vec{s}$$

Proof:

Consider Maxwell's 4th Equation in differential form

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$$

Take surface integration both sides

$$\iint_S \vec{\nabla} \times \vec{H} \, d\vec{s} = \iint_S \left(\vec{J} + \frac{d\vec{D}}{dt} \right) d\vec{s}$$

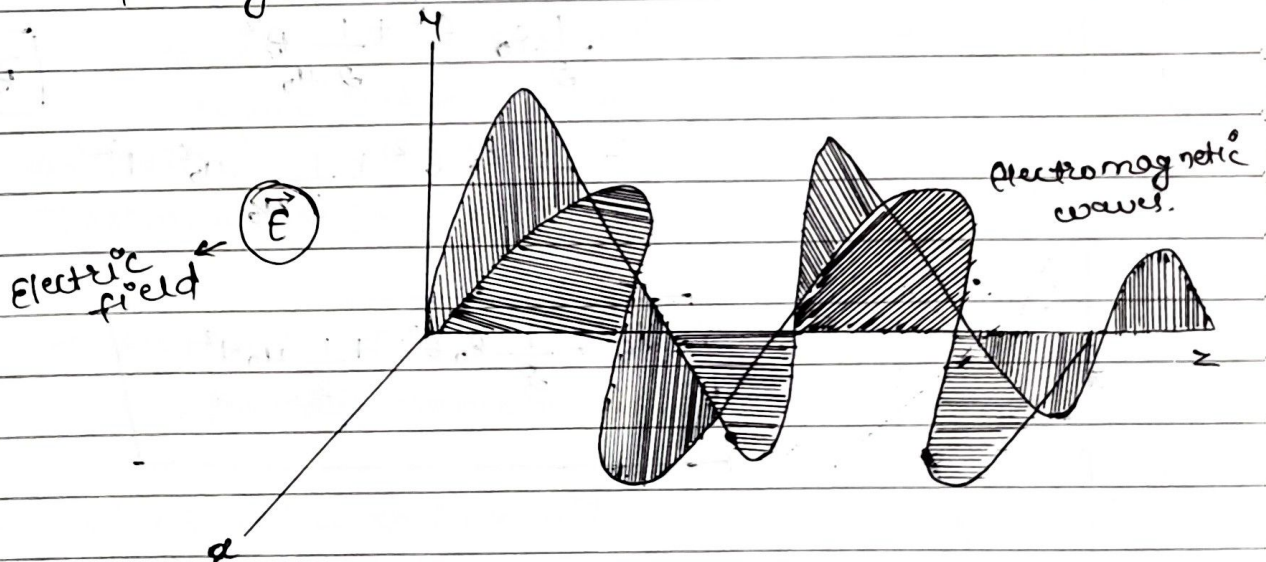
using Stoke's Theorem in L.H.S

$$\oint_L \vec{H} \cdot d\vec{u} = \iint_S \left(\vec{J} + \frac{d\vec{D}}{dt} \right) d\vec{s}$$

Hence Proved.

Electromagnetic waves (EM waves) Electromagnetic Energy

→ Electromagnetic wave are the waves that are created as a result of vibrations between electric and magnetic field.



102
8 marks

Energy stored in electromagnetic waves or
electromagnetic energy.

Energy stored in electric field

$$U_E = \frac{1}{2} \epsilon_0 E^2$$

Energy stored in magnetic field

$$U_B = \frac{1}{2} \mu_0 B^2$$

(Example: μ_0^2 is constant, so it can be taken out of the integral)

Total Energy stored in electromagnetic waves

$$U_{EM} = U_E + U_B$$

$$= \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} \mu_0 B^2$$

$$\boxed{\vec{B} = \mu_0 \vec{H}}$$

$$= \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} \mu_0^2 H^2$$

$$\boxed{U_E = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} \mu_0 H^2}$$

6 Marks
Agile session
mon 20/09/20

Poynting's Theorem

The Poynting theorem says that the work done on the charges by electromagnetic waves is equal to decrease in the energy stored in the field, and minus of energy that thrown out to the surface.

Proof:

consider Maxwell's fourth equation in differential form.

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt} \quad \text{--- (1)}$$

Take dot product with $\vec{E}$ both sides

$$\vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \vec{E} \cdot \left(\vec{J} + \frac{d\vec{D}}{dt} \right) \quad \text{--- (2)}$$

~~$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$$~~

Using standard vector identity

~~$$\vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{H})$$~~

To Remember :- $\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{H})$

~~$$\Rightarrow \vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{H})$$~~

Put this value in eq (2)

$$\vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \left(\vec{J} + \frac{d\vec{D}}{dt} \right) \quad \text{--- (3)}$$

Since $\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$

$$\boxed{\vec{D} = \epsilon_0 \vec{E}}$$

Put this eq in (3)

$$\vec{H} \cdot \left(-\frac{d\vec{B}}{dt} \right) - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{d}{dt} (\epsilon_0 \vec{E})$$

$$\vec{B} = \mu_0 \vec{H} \Rightarrow \vec{H} = \frac{\vec{B}}{\mu_0}$$

$$\frac{\vec{B}}{\mu_0} \cdot \left(-\frac{d\vec{B}}{dt} \right) - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{d}{dt} (\epsilon_0 \vec{E})$$

$$-\frac{\vec{B}}{\mu_0} \cdot \frac{d\vec{B}}{dt} - \vec{E} \cdot \frac{d}{dt} (\epsilon_0 \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J}$$

$$-\frac{d}{dt} \left(\frac{1}{2\mu_0} \vec{B}^2 \right) - \frac{d}{dt} \left(\frac{1}{2} \epsilon_0 \vec{E}^2 \right) - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J}$$

$$\therefore \frac{d}{dt} \left(\frac{1}{2\mu_0} B^2 \right) = \frac{1}{\cancel{2}\mu_0} \cancel{2} B d\vec{B} = \frac{1}{\mu_0} B d\vec{B}$$

$$\frac{d}{dt} \left(\frac{1}{2} \epsilon_0 \vec{E}^2 \right) = \frac{1}{\cancel{2}} \epsilon_0 \cancel{2} E d\vec{E} = \epsilon_0 E d\vec{E}$$

$$-\frac{d}{dt} \left[\frac{1}{2\mu_0} B^2 + \frac{1}{2} \epsilon_0 E^2 \right] - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J}$$

$$\iiint_V \vec{E} \cdot \vec{J} dV = -\frac{d}{dt} \iiint_V \left(\frac{1}{2\mu_0} B^2 + \frac{1}{2} \epsilon_0 E^2 \right) dV - \iiint_V \vec{\nabla} \cdot (\vec{E} \times \vec{H}) dV \quad (4)$$

using Gauss divergence theorem in last term of R.H.S

$$\iiint_V \vec{\nabla} \cdot (\vec{E} \times \vec{H}) dv = \oint_S (\vec{E} \times \vec{H}) \cdot d\vec{s}$$

Put this equation in (4) :

$$\iiint_V \vec{E} \cdot \vec{J} dv = -\frac{d}{dt} \iiint_V \left(\frac{1}{2\mu_0} B^2 + \frac{1}{2} \epsilon_0 E^2 \right) dv - \oint_S (\vec{E} \times \vec{H}) \cdot d\vec{s}$$

The left hand side represents the work done on the charges by electromagnetic field. The first term in R.H.S represents the negative time derivative energy stored in electromagnetic waves and the second term represents the energy carried by electromagnetic waves during its outward flow.

UoVof

1st & 2nd Maxwell's

Poynting vector ($\vec{S}$)

$$\vec{S} = \vec{E} \times \vec{H}$$

$\vec{E}$ = Electric field

$\vec{H}$ = Magnetic field, $\vec{B} = \mu_0 \vec{H}$

→ Poynting vector tells us the rate of flow of electromagnetic waves per unit area per unit time.

The wave Equation of Electromagnetic in free space.

→ In free space Maxwell's four equation are:

(i) $\vec{\nabla} \cdot \vec{D} = 0$ ----- (1) $\rho = 0$

(ii) $\vec{\nabla} \cdot \vec{B} = 0$ ----- (2) $\vec{D} = \epsilon_0 \vec{E}$ $\vec{J} = 0$

$$\vec{B} = \mu_0 \vec{H}$$

$$(iii) \vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt} \quad (3)$$

$$(iv) \vec{\nabla} \times \vec{H} = \frac{d\vec{D}}{dt} \quad (4)$$

Wave Eqn for electric field

Maxwell's third equation is

$$\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$$

Take curl both sides

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla} \times \left(-\frac{d\vec{B}}{dt} \right) \quad (5)$$

Using vector triple product

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - (\vec{A} \cdot \vec{B})\vec{C}$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - (\vec{\nabla} \cdot \vec{\nabla})\vec{E} \quad (6)$$

Since,

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \cdot (\epsilon_0 \vec{E}) = 0$$

$$\vec{\nabla} \cdot \vec{E} = 0$$

$$\vec{\nabla} \cdot \vec{\nabla} = \nabla^2$$

Put these values in eqn (5) (6)

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = 0 - \nabla^2 \vec{E}$$

$$= -\nabla^2 \vec{E}$$

So,

eqn (5) becomes

$$-\nabla^2 \vec{E} = \vec{\nabla} \times \left(-\frac{d\vec{B}}{dt} \right)$$

$$= -\frac{d}{dt} (\vec{\nabla} \times \vec{B})$$

$$\text{Put } \vec{B} = \mu_0 \vec{H}$$

$$-\nabla^2 \vec{E} = -\frac{d}{dt} (\vec{\nabla} \times \mu_0 \vec{H})$$

$$\nabla^2 \vec{E} = \mu_0 \frac{d}{dt} (\vec{\nabla} \times \vec{H})$$

$$\text{Put } \vec{\nabla} \times \vec{H} = \frac{d\vec{D}}{dt} \quad (\text{Maxwell's fourth eqn in free space})$$

$$\nabla^2 \vec{E} = \mu_0 \frac{d}{dt} \left(\frac{d\vec{D}}{dt} \right) \quad \vec{B} = \mu_0 \vec{H}$$

$$\vec{D} = \epsilon_0 \vec{E}$$

$$\nabla^2 \vec{E} = \mu_0 \frac{d^2}{dt^2} (\epsilon_0 \vec{E})$$

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{d^2 \vec{E}}{dt^2}$$

Since,

$$\text{Speed of light } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\Rightarrow c^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$\Rightarrow \frac{1}{c^2} = \frac{1}{\mu_0 \epsilon_0}$$

Put this value in eq (7)

$$\nabla^2 E = \frac{1}{c^2} \frac{d^2 E}{dt^2}$$

This is EM waves equation for electric field.

Similarly, for magnetic field wave equation is

$$\nabla^2 H = \frac{1}{c^2} \frac{d^2 H}{dt^2}$$

Plane electromagnetic waves in free space/vacuum.

(Prove that electromagnetic waves are transverse in nature)

Consider a plane electromagnetic wave travelling along z-axis only.

for such a wave electric field and magnetic vectors are represented by

$$\vec{E} = \vec{E}_0 e^{i(\omega t - kz)} \quad \text{--- (1)}$$

$$\vec{H} = \vec{H}_0 e^{i(\omega t - kz)} \quad \text{--- (2)}$$

$$\vec{\nabla} = \hat{i} \frac{d}{dx} + \hat{j} \frac{d}{dy} + \hat{k} \frac{d}{dz}$$

for z direction,

$$\vec{\nabla} = \hat{k} \frac{d}{dz} \quad \text{--- (3)}$$

$$\nabla^2 = \vec{\nabla} \cdot \vec{\nabla}$$

$$\nabla^2 = \frac{d^2}{dz^2} \quad (4)$$

Now, the wave eq.ⁿ for electromagnetic waves is

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{d^2 \vec{E}}{dt^2} \quad (5)$$

$$\nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{d^2 \vec{H}}{dt^2} \quad (6)$$

Take derivative of (1) w.r.t 'z'

$$\frac{d\vec{E}}{dz} = -ik \epsilon_0 e^{i(\omega t - kz)}$$

$$\frac{d\vec{E}}{dz} = -ik \vec{E}$$

$$\boxed{\frac{d}{dz} = -ik} \quad (7)$$

Now, take derivative of (1) w.r.t 't'

$$\frac{d\vec{E}}{dt} = i\omega \epsilon_0 e^{i(\omega t - kz)}$$

$$\frac{d\vec{E}}{dt} = i\omega \vec{E}$$

$$\boxed{\frac{d}{dt} = i\omega} \quad (8)$$

Now Maxwell's first and second eqⁿ in free space ($\rho=0$) are

$$\vec{\nabla} \cdot \vec{D} = 0$$

$$\left(\hat{k} \frac{d}{dz} \right) \cdot (\epsilon_0 \vec{E}) = 0$$

$$\Rightarrow \hat{k} \frac{d}{dz} \cdot \vec{E} = 0$$

using eq (1)

$$\Rightarrow \hat{k} (-i k) \cdot \vec{E} = 0$$

$$\Rightarrow -i k (\hat{k} \cdot \vec{E}) = 0$$

$$\Rightarrow \boxed{\hat{k} \cdot \vec{E} = 0}$$

$\hat{k}$ = direction of electromagnetic

Hence EM wave and electric field $\vec{E}$ for
 $\perp$ to each other.

Again,

for Maxwell second eqⁿ.

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \cdot (\mu_0 \vec{H}) = 0$$

$$\Rightarrow \vec{\nabla} \cdot \vec{H} = 0$$

$$\hat{k} \frac{d}{dz} \cdot \vec{H} = 0$$

using eq (2)

$$\hat{k} (-i k) \cdot \vec{H} = 0$$

$$-i k (\hat{k} \cdot \vec{H}) = 0$$

$$\Rightarrow \boxed{\hat{k} \cdot \vec{H} = 0}$$

EM waves and Magnetic field $\vec{B}$ are $\perp$ to each other.